LUX PROVIDE

EXPLORING THE FRONTIERS OF DIGITAL INTELLIGENCE

MADE IN LUXEMBOURG

Profiling Applications and Code on High-Performance Computing (HPC)

Training Agenda

[13h30 – 13h40] Outline of the training

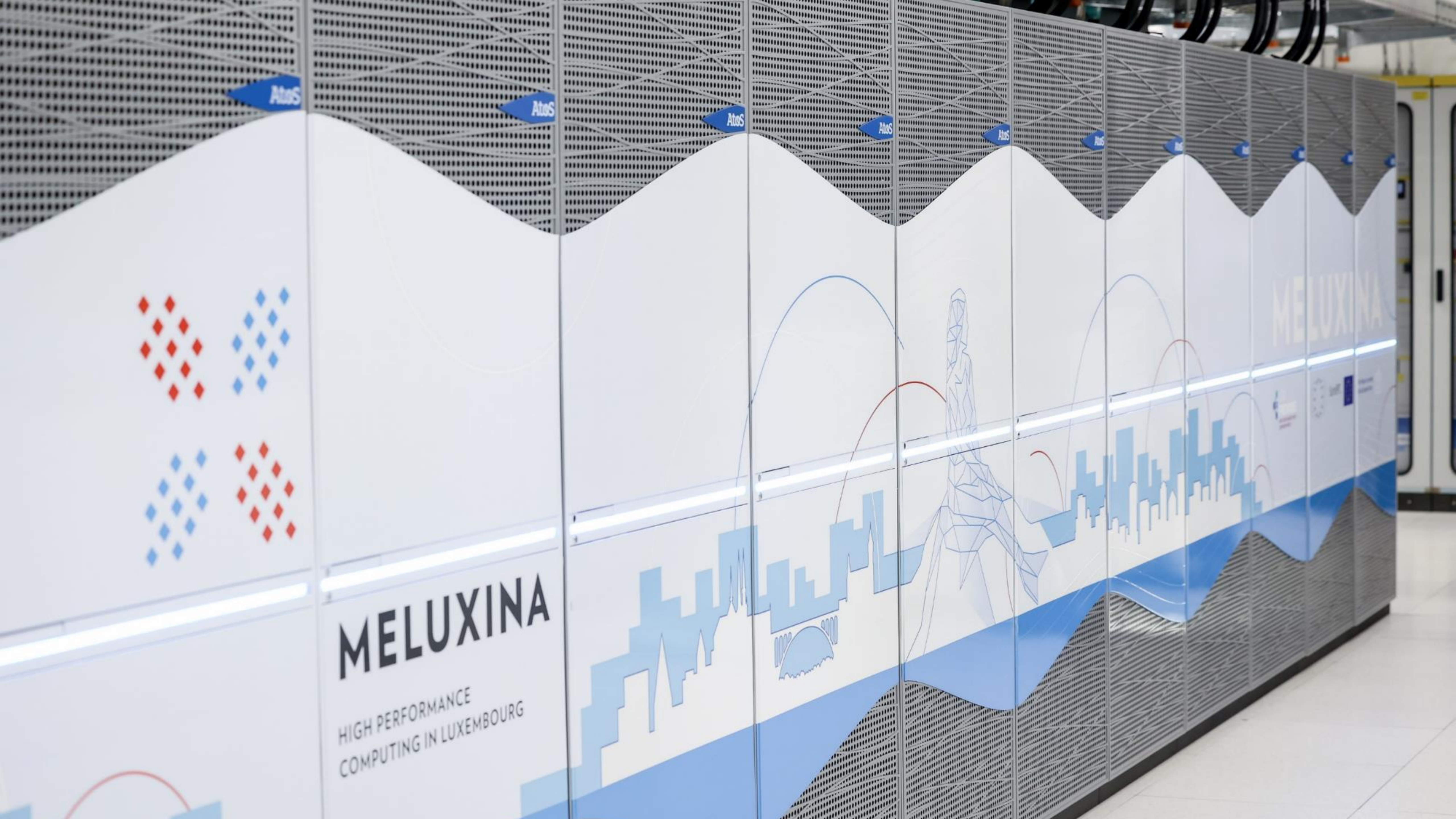
[13h40 – 14h00] Connecting to the machine (MeluXina)

[14h00 – 14h30] Code optimization based on profiling: an inefficient training loop in PyTorch

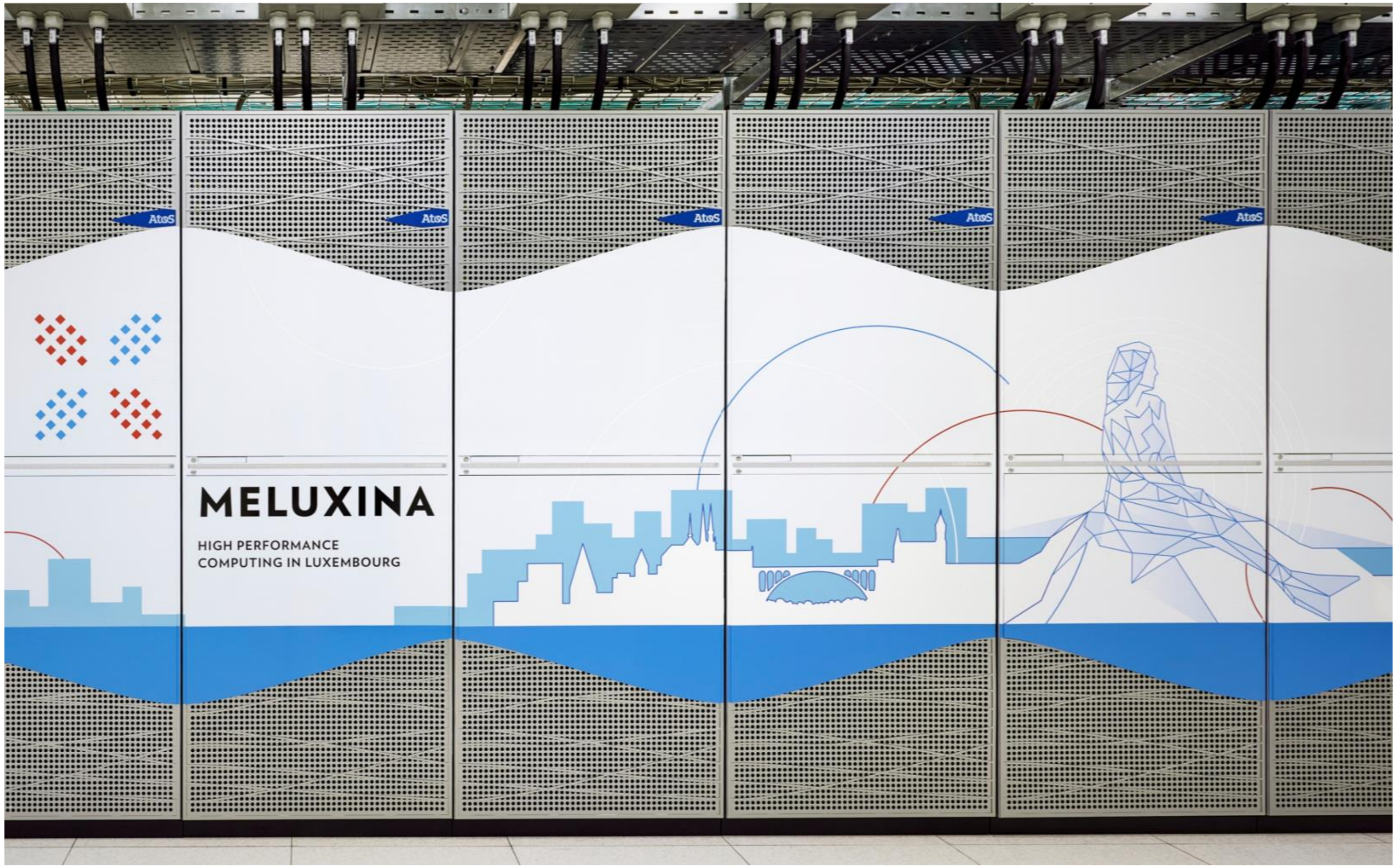
[14h30 – 14h55] Hands on & Contest

[14h55 – 15h00] Contest results

MeluXina


Atos
MELUXINA
HIGH PERFORMANCE
COMPUTING IN LUXEMBOURG
MELUXINA

LUXEMBOURG’S NATIONAL SUPERCOMPUTER MELUXINA



90.000
HPC CPU cores

800
GPU-AI accelerators

20
Petabytes high performance storage

+450
TB RAM

400+
Tailored software packages

Internal chat & slides

https://tlk.io/scynergy_optimization

https://github.com/LuxProvide/scynergy_profiling_training

Connecting to the machine (MeluXina)

ACCESSING MELUXINA

- **ssh *user@login.lxp.lu* –p 8822**
  - 4 login nodes in total, at the moment round-robin between 2 login nodes
  - Login nodes directly accessible (not recommended)
  - Can switch between login nodes
- **Run the following command (once connected)**
  - *cd /project/scratch/p200865*
  - *cd 0_BasicTest/*
  - *sbatch basic_launcher.sh*
- **Check outputs**
  - *cd /project/scratch/p200865/output/${USER}*

```
Welcome the Luxembourg - EuroHPC supercomputer
MeluXina

| You are on a MeluXina login node                                               |
|                          System information: Compute                           |
| Nodes | CPU                       | RAM     | Accelerator              | Disk  |
| 573N  | 2x AMD 7H12: 128c @2.6G   | 512GB   | -                        | -     |
| 200N  | 2x AMD 7452:  64c @2.3G   | 512GB   | 4x NVIDIA A100-40        | 1.92G |
|  20N  | 2x AMD 7452:  64c @2.3G   | 512GB   | 2x Intel Stratix 10MX-16 | 1.92G |
|  20N  | 2x AMD 7H12: 128c @2.6G   | 4096GB  | -                        | 1.92G |
|                          System information: Data                              |
| Tier          | Capacity | Speed    | Type | Location on compute/login        |
| Scratch       | 0.6PB    | 400GB/s  | NVMe | /project/scratch                 |
| Home/Project  | 12.5PB   | 180GB/s  | HDD  | /home/users, /project/home       |
| Backup        | 7.5PB    | 30GB/s   | HDD  | -                                |
|                          System information: Interconnect                      |
| Fabric: Infiniband HDR, 200Gbps, DragonFly+ topology                           |
| Links : 1x on CPU nodes, 2x on GPU, FPGA & LargeMemory nodes                   |
|                          System information: Software                          |
| Production software stack: 2021.3, current default and obsolete on             |
| 14 Jan 2023                                                                    |
|                                                                                |
| Production software stack: 2022.1, default from 15 Jan 2023, use it            |
| now with: `module load env/release/2022.1`                                     |
|                                                                                |
|                          Center information                                    |
| News & Events        : luxprovide.lu                                           |
| Documentation        : docs.lxp.lu                                             |
| System status        : weather.lxp.lu                                          |
| Support              : servicedesk.lxp.lu, servicedesk@lxp.lu                  |
| LinkedIn & Twitter : @luxprovide #meluxina @EuroHPC_JU                         |

##################################################################################
#                    System events, in-progress and upcoming                     #
#--------------------------------------------------------------------------------#
# * 2023-10-30 09:00 - 19:00 CET MeluXina update for performance, security      #
#   and reliability. Contact our Service Desk in case of any observed issues.    #
##################################################################################
Last login: Thu Nov 16 17:01:41 2023 from 158.64.12.102
[fbongiovanni@login02 ~]$
```

Code optimization based on profiling: An inefficient training loop in PyTorch

Code Optimization based on profiling

Why profiling? The main reasons

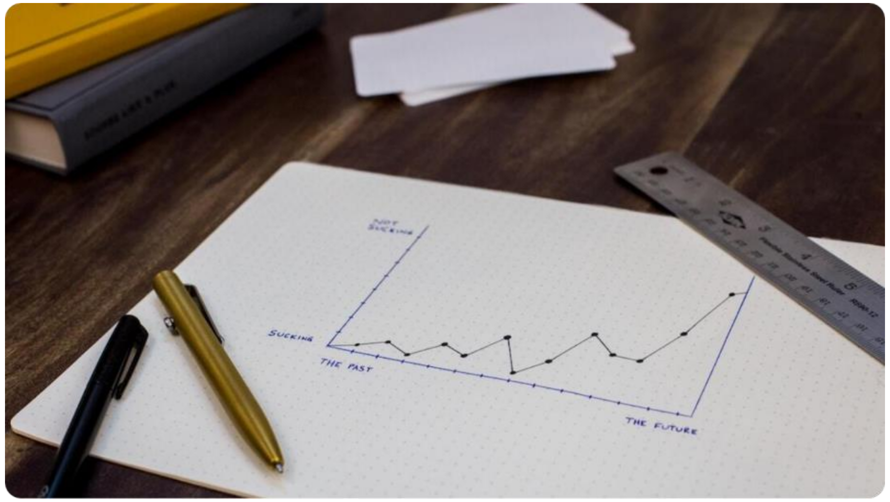
•**Identify Bottlenecks**: where your code spends the most time? Find the stages (CPU, GPU, memory, I/O) .

•**Optimize Resource Utilization** Ensure full use of expensive resources (GPU cores, memory bandwidth, tensor cores, etc.).

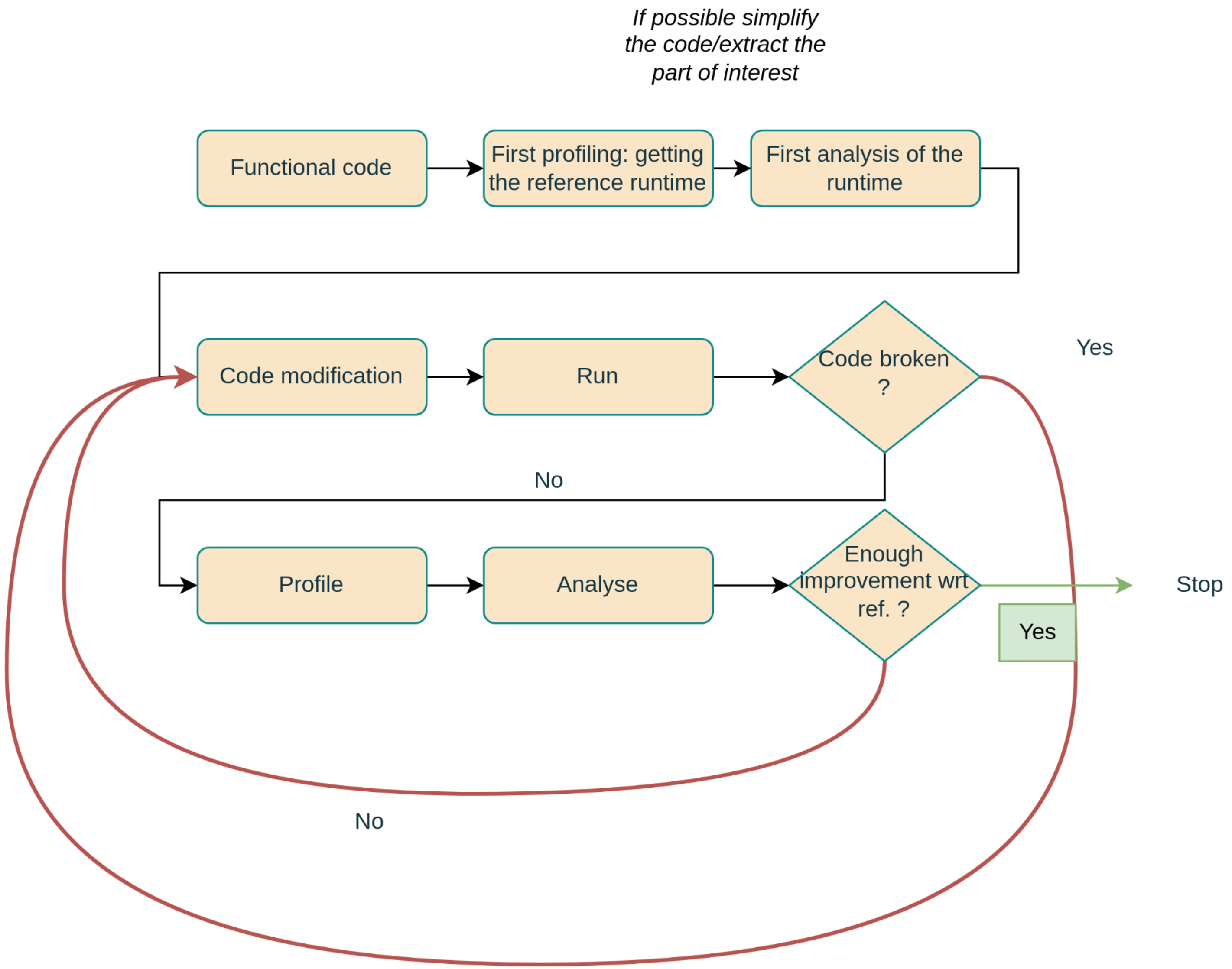
•**Minimize Data Transfer Overhead:** Detect slowdowns caused by unnecessary CPU↔GPU or GPU↔GPU data transfers.

•**Reduce Latency and Increase Throughput**: Profile kernels to speed up individual operations and improve batch processing.

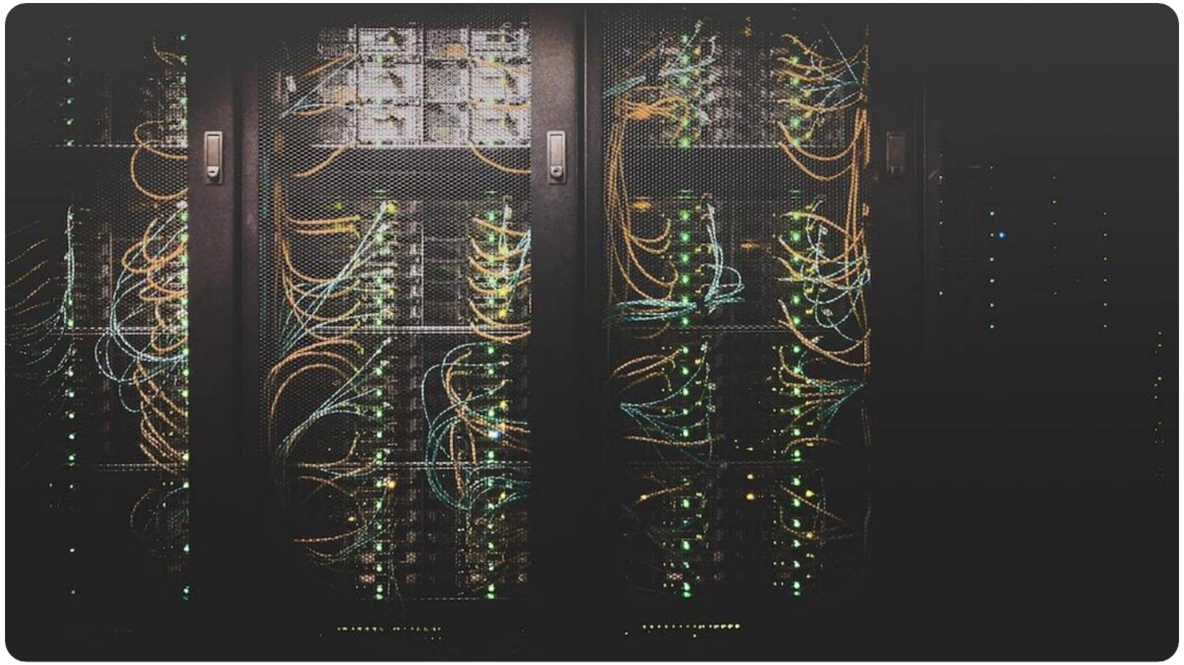
•**Scalability Analysis:** Understand how performance scales with model/data size or number of accelerators.



Profiling: The standard workflow



A first example: Let's profile

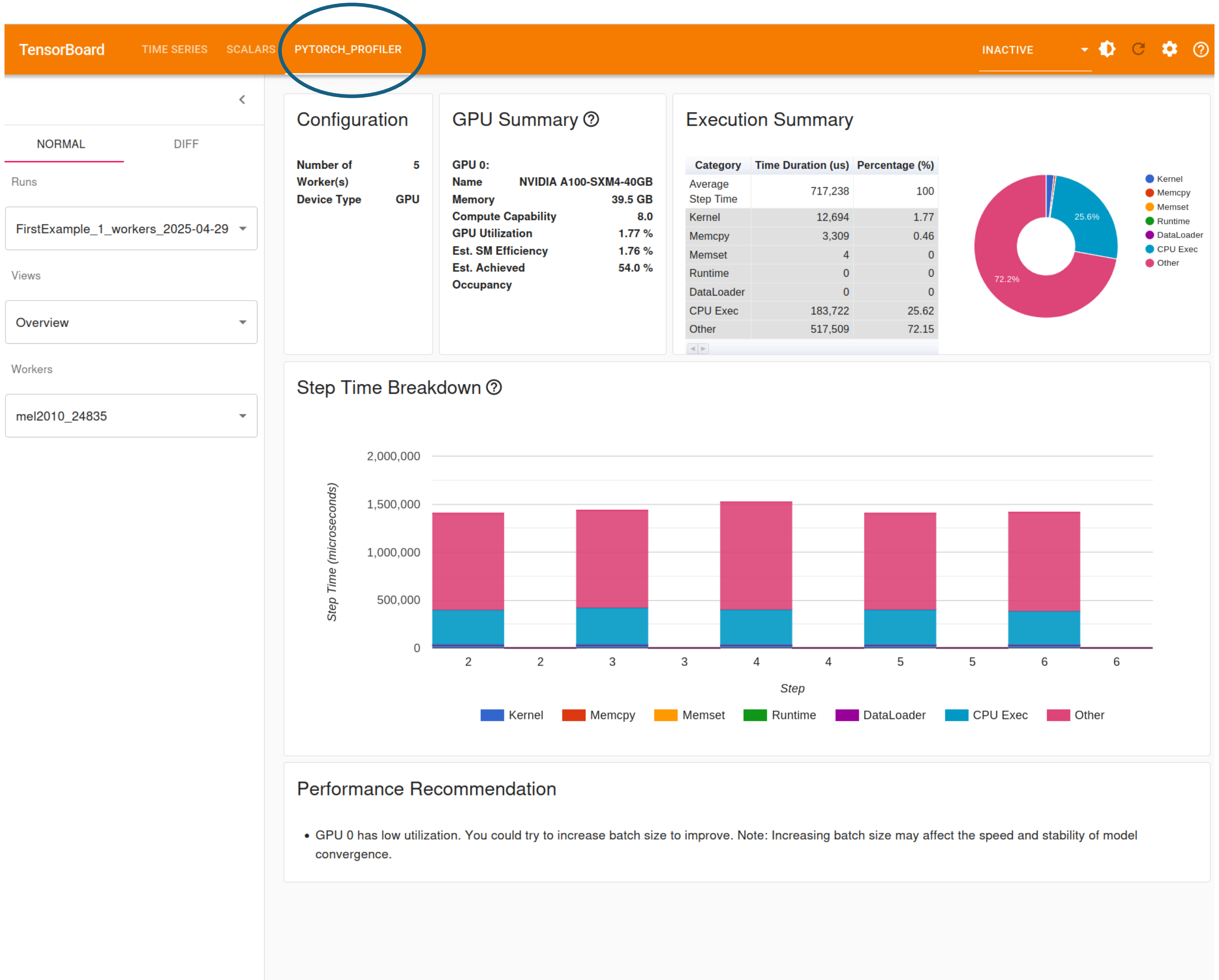


- **Allocate resources on the machine**

  - cd /project/scratch/p200865/
  - source get_interactive_job_for_profiling.sh
  - cd /project/scratch/p200865/1_ExampleMnist
  - source launcher.sh

The launcher.sh will be modified several time during the training.

A first example: Focus on the torch profiler



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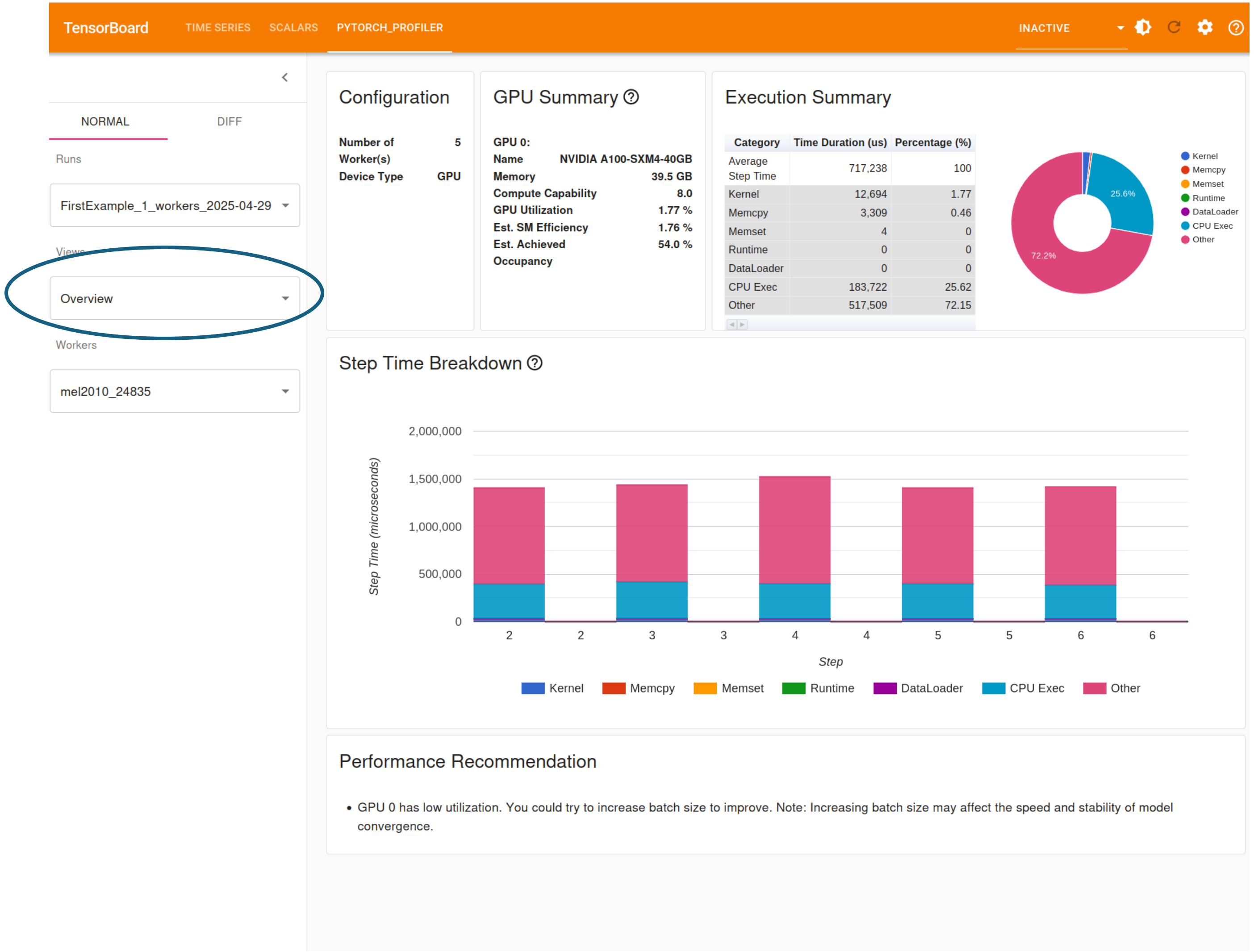
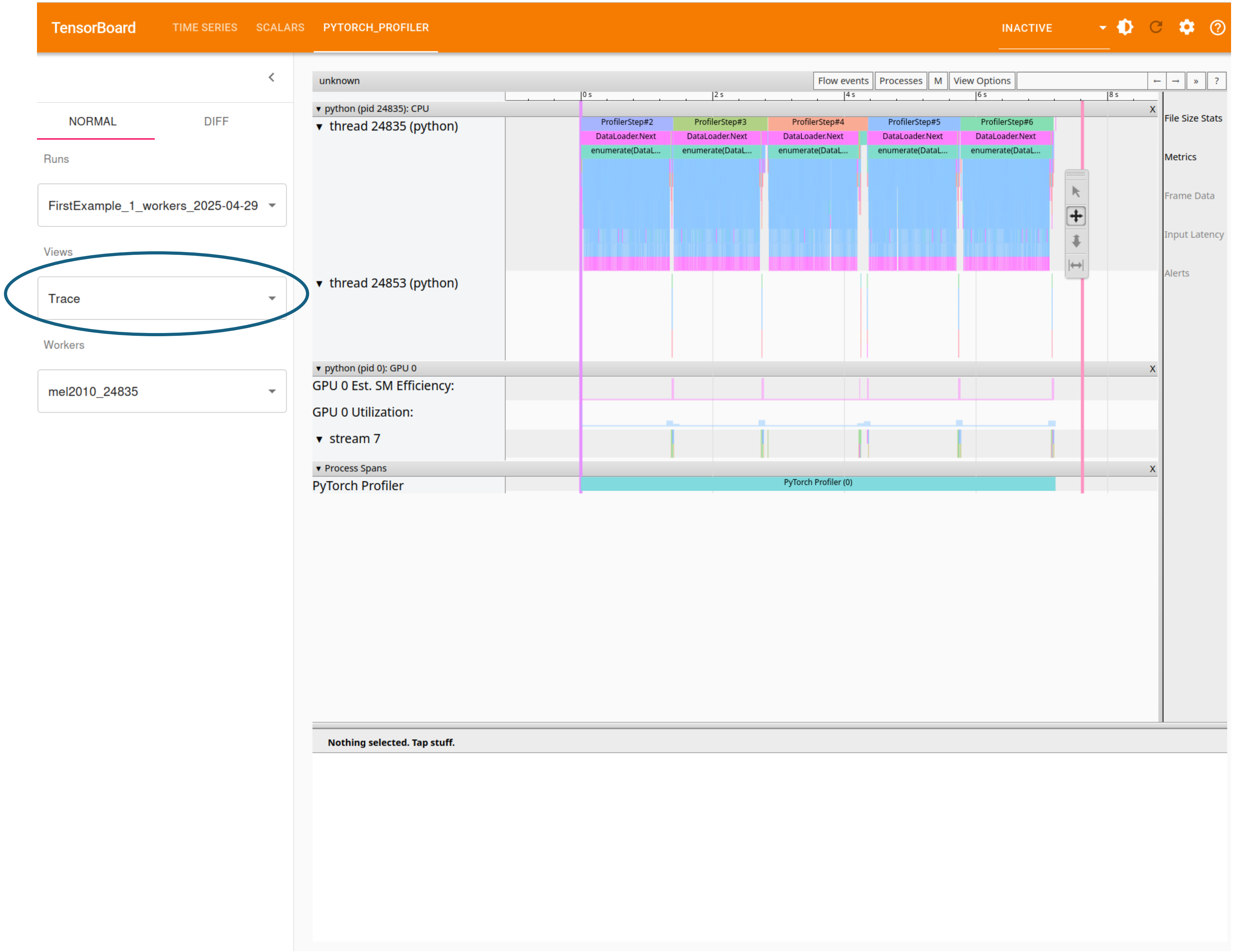
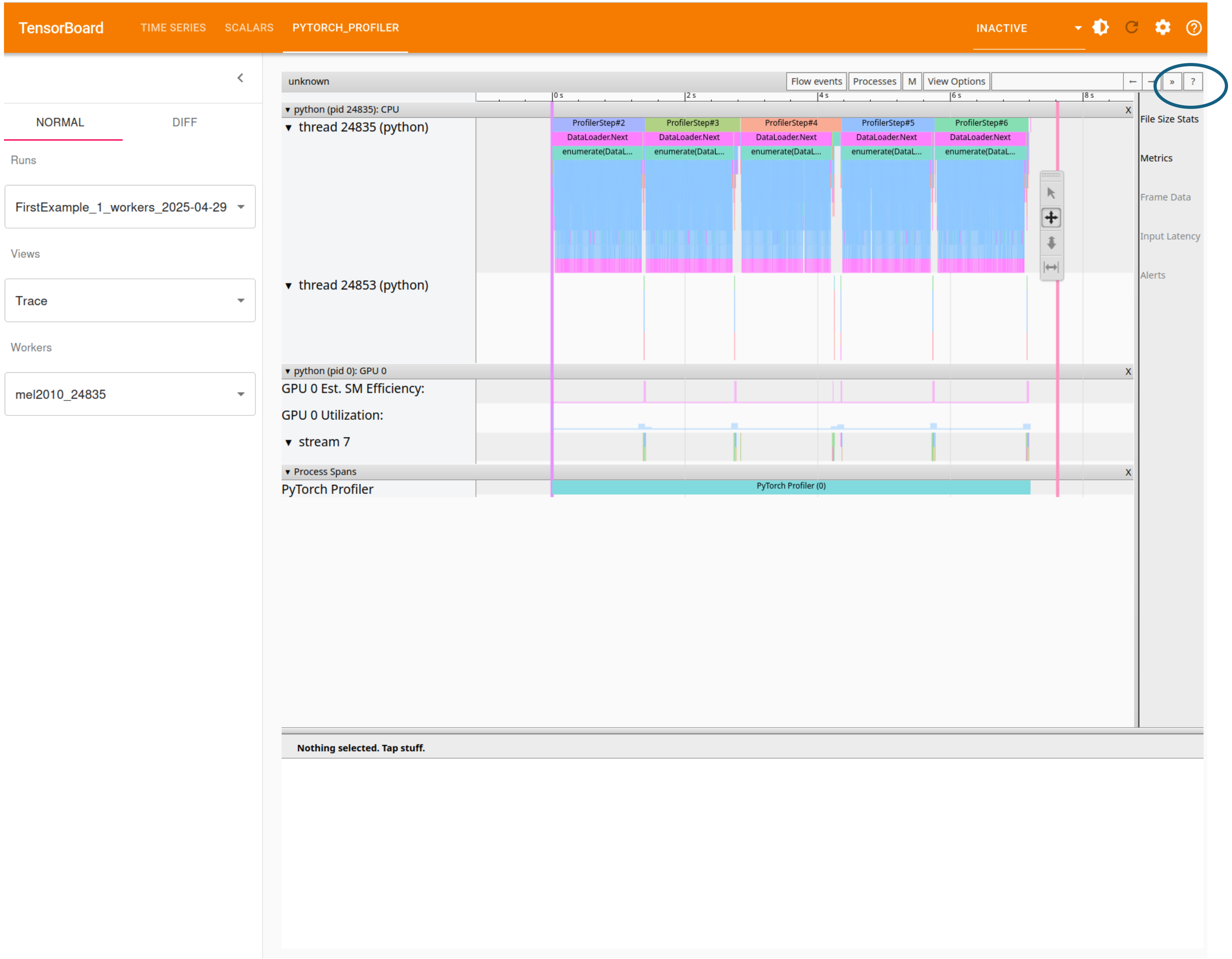


A first example: Focus on the torch profiler

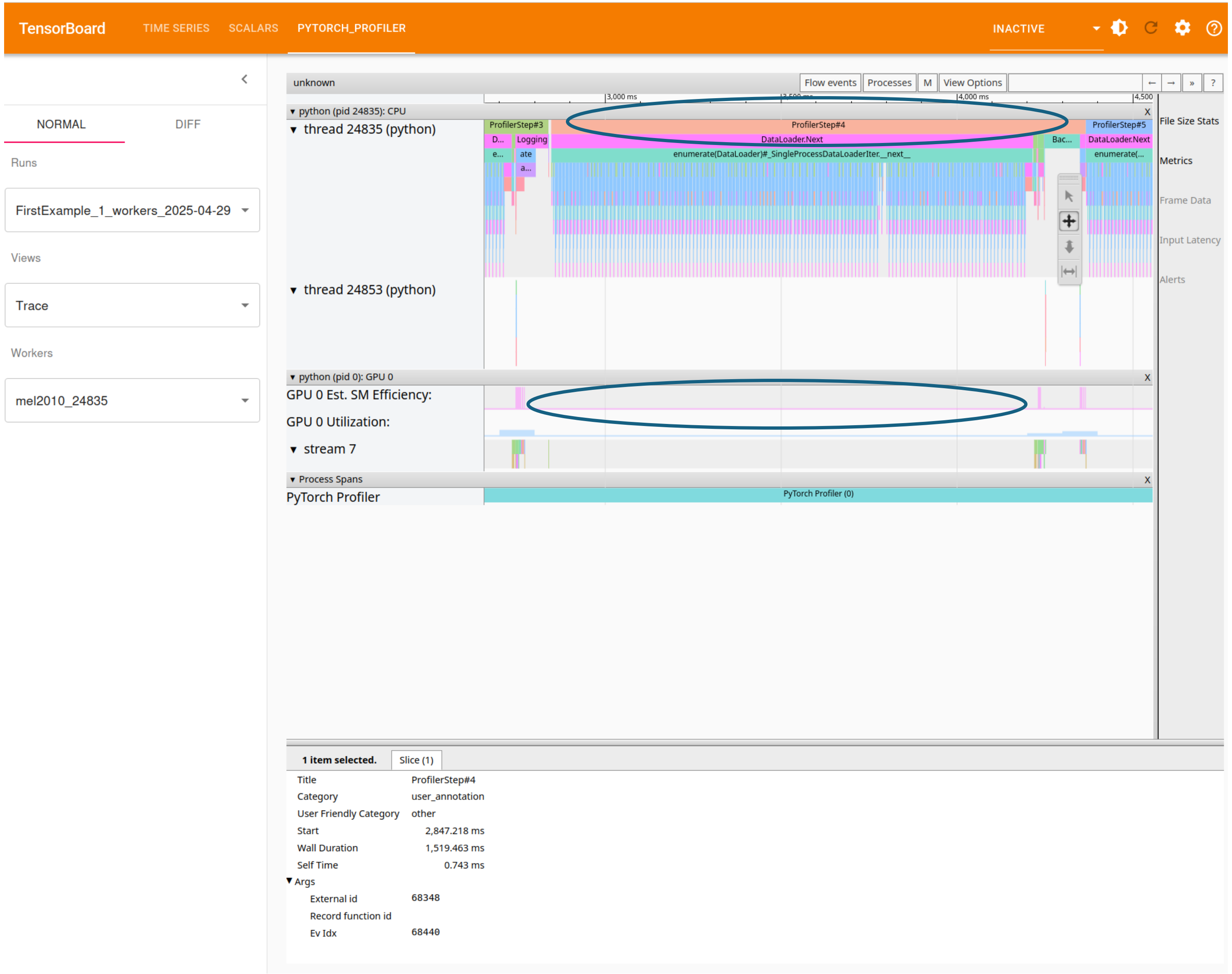


A first example: Focus on the torch profiler



All the shortcuts are Explained here

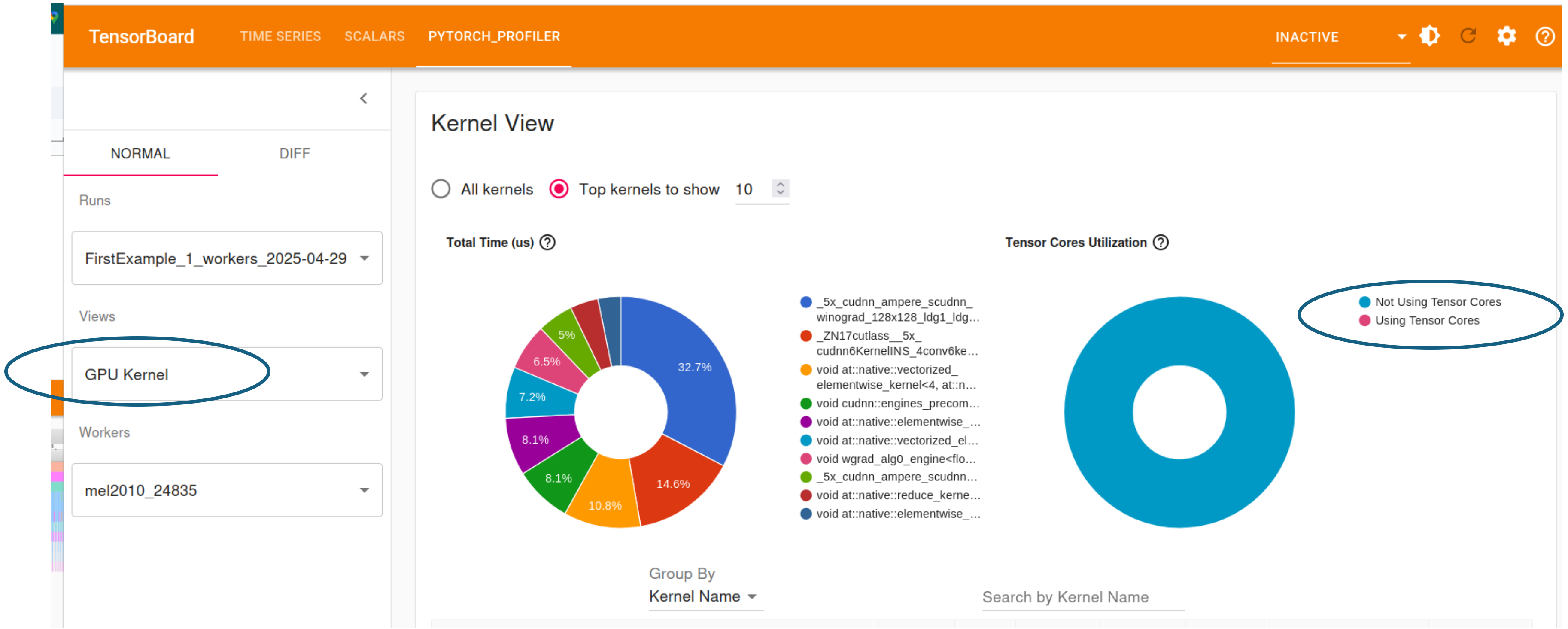
A first example: Focus on the torch profiler



Click on the profiler step 4 and press "f" to focus on this region

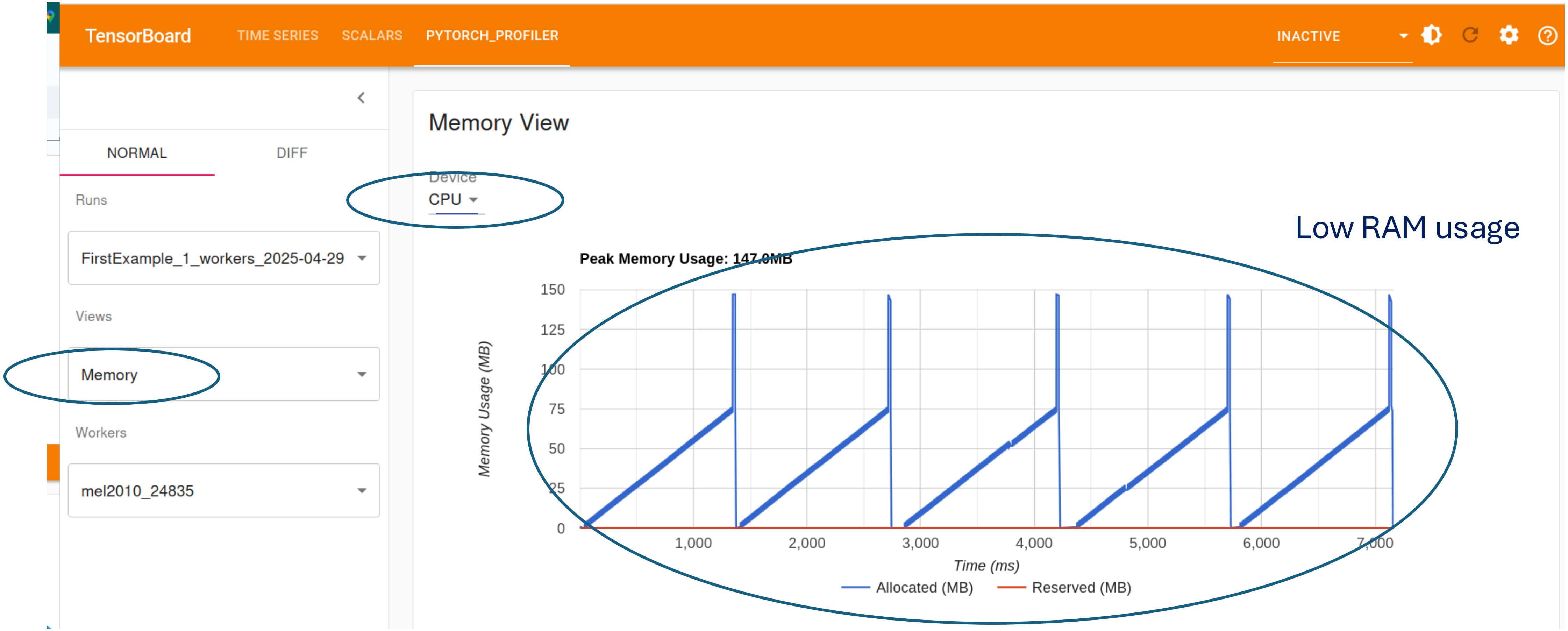
Low GPU usage :(

A first example: Focus on the torch profiler

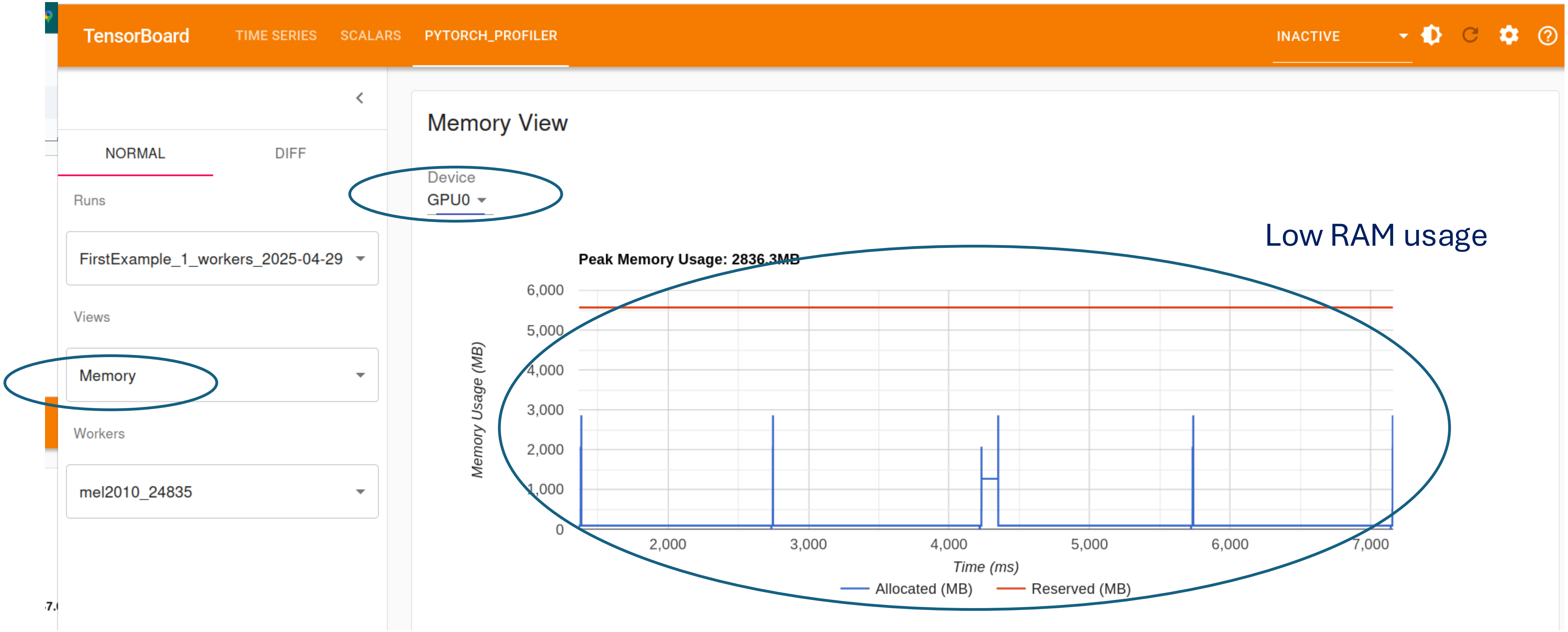


Not using tensor cores

A first example: Focus on the torch profiler



A first example: Focus on the torch profiler

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